

Composite three-dimensional + drainage method

Unit 1/4 · 3D shapeadvanced · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Units 1 and 4 + Modern Education 5B Unit 1 and 4

Core traps: □ T1 Composite solid - double counting of overlapping parts · T4 Drainage method - rising water level vs overflow confusion

SSPA association: ● High frequency Commonly used in paper two, accounting for about 8-10%

Prerequisite knowledge: Class 26 (Volume and surface area of cuboid/cube) · P4 Basic concepts of volume

Objective of this class: ❶ Calculate the volume by composite solid segmentation method ❷ Calculate the volume of irregular objects by drainage method ❸ Overflow problem ❹ Comprehensive application

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Cuboid $8 \times 6 \times 4$ cm. Volume = ?	Basic	
2	The side length of the cube is 5 cm. Volume = ?	Basic	
3	A rectangular water tank is $20 \times 15 \times 10$ cm. How many cm^3 can it hold at most?	Basic	
4	Calculation: $12^3 = ?$ (i.e. $12 \times 12 \times 12$)	Advanced	
5	How would you calculate the volume of an L-shaped solid (consisting of two cuboids)? Write out your approach.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Volume of a composite solid (division method/complementation method)

● SSPA

Compound solid = a shape formed by the combination of two or more simple solids. Two solution methods:

- ① **Segmentation method (recommended) |||SEP|||**: Cut the composite solid into several cuboids/cubes → find the volumes separately → add up
Complementary method |||SEP|||: Fill the missing corners into a complete cuboid → Find the larger volume → Subtract the supplemented part
- ② **Key: Do not repeatedly calculate the overlapping parts!** ④ Steps: draw auxiliary lines → mark the dimensions of each segment → calculate by segments → add up
- ③ **Key: Don't double-count overlapping parts!**
- ④ Steps: Draw auxiliary lines → mark the dimensions of each segment → calculate by segments → add up

□ Trap detonation example (most common mistakes)

The picture below shows an L-shaped composite volume: bottom layer $10 \times 6 \times 3$ cm, upper layer on the right $4 \times 6 \times 4$ cm. Total volume = ?

✗ Common mistakes (55% students)

Treat it all as a large cuboid:

$$V = 10 \times 6 \times 7 = 420 \text{ cm}^3 \quad \text{✗}$$

The L shape is not a complete cuboid, the upper left corner is empty! Multiplying directly will overcount the empty part.

✓ Correct solution (segmentation method)

$$\text{Ground floor: } 10 \times 6 \times 3 = 180 \text{ cm}^3$$

$$\text{Upper layer: } 4 \times 6 \times 4 = 96 \text{ cm}^3$$

$$\text{Total} = 180 + 96 = 276 \text{ cm}^3$$

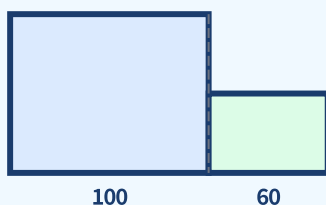
Cut into two rectangles, calculate each and then add up.

 **Tip: "A composite solid needs to be divided. Draw a line to cut the canal. Calculate the volume of each piece clearly. Add and bury it to determine the total volume!"**

⚠ The most frequent error: treating the composite solid as a complete cuboid and directly multiplying it by the peripheral dimensions.

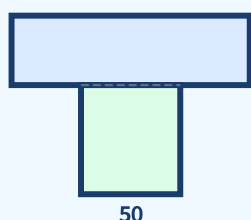
⚠ The second most frequent error: the overlapping part is counted twice after segmentation. The dividing lines must be at the border and not overlap.

◆ L-shaped compound solid - example of segmentation method



Bottom $w_1 \times h_1$ + right upper layer $w_2 \times h_2$. The dividing line (gray dotted line) cuts the L shape into two cuboids, and the volumes are calculated separately and then added.

◆ T-shaped composite three-dimensional



Lower horizontal bar $w_{\text{top}} \times h_{\text{top}}$ + upper vertical bar $w_{\text{bot}} \times h_{\text{bot}}$. Use the same dividing method: cut into upper and lower pieces and count them separately.

□ Hollowing out a solid - example of complementation method



Complete cuboid – removed solid = remaining volume. Complementary method: first calculate the large cuboid, and then subtract the hollowed out part.

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	T-shaped composite three-dimensional: lower layer $12 \times 8 \times 3$ cm, upper layer (standing in the middle) $6 \times 8 \times 5$ cm. Total volume = ? (Picture segmentation calculation)	🌱	
Example 2	Ladder-shaped three-dimensional (three layers): bottom layer $10 \times 5 \times 2$ cm, middle layer $8 \times 5 \times 2$ cm, top layer $6 \times 5 \times 2$ cm. Total volume = ?	🌱	
Example 3	Use the complement method to find: A cuboid is $8 \times 6 \times 5$ cm, and the upper right corner is cut off ($3 \times 3 \times 2$ cm). Remaining volume = ?	🌳	

Knowledge Point 1 Synchronization practice

#	Question	Difficulty	Working Space
6	L-shaped solid: bottom layer $8 \times 5 \times 2$ cm, upper layer on the right side $3 \times 5 \times 3$ cm. Total volume = ?	🌱	
7	T-shaped three-dimensional: lower width $14 \times 6 \times 3$ cm, upper vertical (center) $4 \times 6 \times 5$ cm. Total volume = ?	🌱	

8	Cuboid $10 \times 8 \times 6$ cm, with one corner cut off ($4 \times 4 \times 3$ cm). Remaining volume = ?		
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Knowledge Point—advanced practice

#	Question	Difficulty	Working Space
9	Cross-shaped composite solid: horizontal cuboid $20 \times 5 \times 3$ cm, vertical cuboid $5 \times 12 \times 5$ cm passing through the center. Total volume = ?		
10	The composite solid is composed of cube A (side length 8 cm) and cube B (side length 5 cm) laminated together (one side is completely laminated). Total volume = ?		
11	Glue 3 identical cubes (side length 4 cm) in a row. The total volume of the composite solid = ? Surface area = ? (Note that the adhesive surface is not included in SA)		

Knowledge point 2: Drainage method - finding the volume of irregular objects SSPA required test

Archimedes' principle (drainage method):

- ① Completely immerse the object in the water → the volume of the rising water level = the volume of the object
- ② **Core formula: volume of the object = bottom area of the container × rising height of the water level**
- ③ That is: $V_{\text{object}} = L_{\text{container}} \times W_{\text{container}} \times (h_{\text{Back}} - h_{\text{Before}})$
- ④ **Prerequisites:** Object must **completely submerged** in water; water cannot overflow

① before immersion Record the original water level height h_1	② fully immersed Objects submerged in water must not come out of the water	③ after immersion Record the new water level height h_2	④ calculate $V = \text{base area} \times (h_2 - h_1)$
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❏ Trap detonation example

The rectangular water tank is $20 \times 15 \times 10$ cm, with an original water depth of 4 cm. After placing a stone (completely submerged), the water level rose to 7 cm. Volume of stone = ?

Common mistakes (40% students)

$$V = 20 \times 15 \times 7 = 2100 \text{ cm}^3 \quad \text{✗}$$

This calculation is "the total volume of water + stone now", not the volume of stone!

Correct solution

$$\text{Base area} = 20 \times 15 = 300 \text{ cm}^2$$

$$\text{Water level rise} = 7 - 4 = 3 \text{ cm}$$

$$\text{Stone volume} = 300 \times 3 = \mathbf{900 \text{ cm}^3}$$

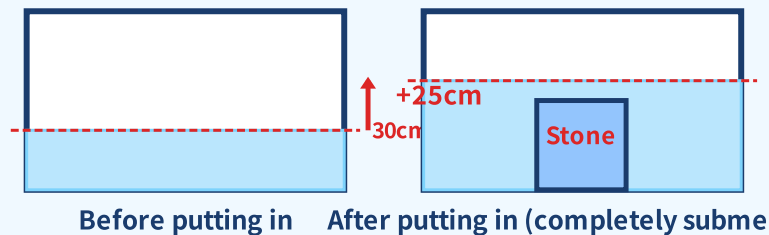
Key: rising "water level difference" × bottom area = volume of the object

 **Tip: "The drainage method is used to find the volume, not the total water volume! Multiply the bottom area by the water level difference, multiply by the amount of the**

rise!"

⚠ Fatal error: Use the "water level after immersion" to directly multiply the bottom area and forget to subtract the "original water level".

💧 Schematic diagram of drainage method — Archimedes' principle



Water surface rise = volume of object ÷ bottom area

Left: water level before input w_b cm. Right: water level w_a cm after input. Water level rise = $w_a - w_b$ cm. Volume of object = bottom area of container $\times$ ($w_a - w_b$). Key: The object must be completely submerged!

🧠 **Three steps of drainage method:** ① Calculate the bottom area (container length $\times$ width) ② Calculate the water level difference (new – old) ③ Volume = bottom area $\times$ water level difference

Knowledge Point 2 Worked Examples

#	Question	Difficulty	Working Space
Example 4	The water tank is $25 \times 10 \times 15$ cm, and the original water depth is 5 cm. After adding the iron block, the water level rose to 9 cm. Volume of iron block = ?	🌱	
Example 5	The bottom area of the water tank is 200 cm^2 , and the original water depth is 6 cm. The water level rose by 4 cm after the object was placed. Object volume = ?	🌿	
Example 6	The water tank is $30 \times 20 \times 20$ cm, and the original water depth is 8 cm. Place an object with a volume of 1200 cm^3 (completely submerged). new water level = ?	🌳	

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
12	The water tank is $15 \times 10 \times 12$ cm, and the original water depth is 3 cm. The water level rose to 8 cm after placing the stones. Volume of stone = ?	🌱	
13	The bottom area of the water tank is 150 cm^2 , and the original water depth is 5 cm. The water level rose by 6 cm after placing the metal block. Volume of metal block = ?	🌿	
14	The water tank is $40 \times 25 \times 30$ cm, and the original water depth is 10 cm. Place an object with a volume of 2000 cm^3 . new water level = ?	🌳	

15	The side length of the cube water tank is 20 cm, and the original water depth is 8 cm. Place a cube of iron with side length 10 cm (completely submerged). new water level = ?		
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Knowledge point 3: Overflow problem (water level exceeds the height of the container) SSPA killer question

When the object placed is too large and the water level exceeds the height of the container:

- ① Calculate first: the theoretical water level after placing the object = original water level + object volume $\div$ bottom area
- ② If the theoretical water level > container height $\rightarrow$ water will overflow
- ③ **Overflow water amount = (theoretical water level – container height) $\times$ bottom area**
- ④ Or: Overflow amount of water = volume of object – (available space in container)
- ⑤ Available space in container = bottom area $\times$ (container height – original water level)

Trap detonation example

The water tank is $20 \times 15 \times 10$ cm, and the original water depth is 6 cm. Place a stone with a volume of 1500 cm^3 (completely submerged). How many cm^3 of water spilled?

Common mistakes (60% students)

Theoretical water level = $6 +$

$$1500 \div (20 \times 15)$$

$$= 6 + 5 = 11 \text{ cm}$$

$11 > 10$, so water spills 1 cm high

$$\text{Overflow} = 20 \times 15 \times 1 = 300 \text{ cm}^3$$

The answer 300 cm^3 ? is actually correct...but

The mistake most students make is: using the drainage method formula without checking whether there is overflow! Or forgot to subtract the free space.

Correct solution (available space method)

$$\text{Available space} = 20 \times 15 \times (10 - 6)$$

$$= 300 \times 4 = 1200 \text{ cm}^3$$

$$1500 > 1200, \text{ so overflow}$$

$$\text{Overflow} = 1500 - 1200$$

$$= 300 \text{ cm}^3$$

Both methods are available. It is more intuitive to use the "available space method" - first calculate how much can be installed.

 **Tip: "First calculate how much available space there is. If the object is too large, it will leak out - how much it will leak out = object volume - available space."**

 **The step most likely to leak: When you see the "overflow problem", you must first check whether the water level exceeds the height of the container! It will be wrong to directly apply the drainage formula without checking it.**

Knowledge Point Three Worked Examples + Synchronous Practice

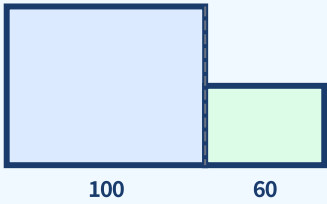
#	Question	Difficulty	Working Space
Example 7	The water tank is $15 \times 10 \times 8$ cm, and the original water depth is 5 cm. Place an object with a volume of 600 cm^3 . Will the water overflow? How much overflow?		

Example 8	Water tank $30 \times 20 \times 15$ cm, water depth 12 cm. Place a cube of iron with a side length of 10 cm. Will the water overflow? How much overflow?		
16	The water tank is $20 \times 10 \times 10$ cm, and the original water depth is 7 cm. Place an object with a volume of 500 cm^3 . Overflow water volume = ?		
17	Water tank $25 \times 20 \times 12$ cm, water depth 10 cm. Place an object with a volume of 800 cm^3 . Overflow water volume = ?		



 3D shape reference pictures (or less practice questions need to refer to these graphics)

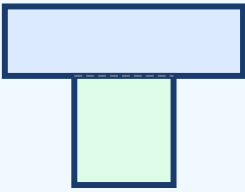
L-shaped composite three-dimensional



100 60

Reference Q1, Q3, Q6, Q18, Q23

T-shaped composite three-dimensional



50

Reference Q7

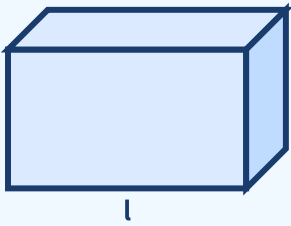
Hollow out the three-dimensional



dig away

Refer to Q3, Q8, Q20

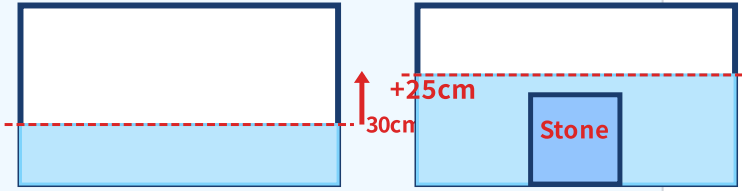
Standard cuboid



l

Length (l) $\times$ width (w) $\times$ height (h). $V=l \times w \times h$. $SA=2(lw+lh+wh)$.

Drainage method standard diagram (complete submersion, rising water level)



Before putting in After putting in (completely submersed)

Water surface rise = volume of object $\div$ bottom area

 Red letters mark water level changes. Bottom area $\times$ water level difference = object volume (not bottom area $\times$ new water level!)

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

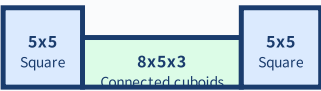
#	Question	Difficulty	Working Space
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18	L-shaped solid: bottom layer $6 \times 4 \times 2$ cm, upper right side $2 \times 4 \times 3$ cm. Total volume = ?		
19	The water tank is $20 \times 10 \times 15$ cm, and the original water depth is 5 cm. The water level rose to 9 cm after placing the stones. Volume of stone = ?		
20	Cuboid $15 \times 10 \times 8$ cm, with one corner cut off ($5 \times 5 \times 4$ cm). Remaining volume = ?		
21	The bottom area of the water tank is 120 cm^2 , and the original water depth is 4 cm. The water level rose by 5 cm after the object was placed. Object volume = ?		
22	Water tank $18 \times 12 \times 10$ cm, water depth 8 cm. Is there room to put another object with a volume of 360 cm^3 without overflowing?		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
23	Compound solid: cube A (side length 6 cm) is placed on top of cube B (side length 4 cm, centered). Total volume = ?		
24	The water tank is $30 \times 20 \times 25$ cm, and the original water depth is 10 cm. Place an object with a volume of 3000 cm^3 . new water level = ? Will the water overflow?		
25	The side length of the cube water tank is 15 cm and the water depth is 10 cm. Place a cube of iron with a side length of 8 cm. new water level = ?		
26	Water tank $24 \times 16 \times 18$ cm, water depth 14 cm. Place a cube with side length 6 cm. Overflow water volume = ?		
27	An empty water tank $20 \times 15 \times 12$ cm. First, 2400 cm^3 of water is injected, and then a stone with a volume of 600 cm^3 is placed. Final water depth = ? Is the water overflowing?		

challenge layer (total 3 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
28	 A U-shaped composite solid (there is a cube on each side, connected by a cuboid in the middle): the left cube has a side length of 5 cm, the right cube has a side length of 5 cm, and the middle connecting cuboid is $8 \times 5 \times 3$ cm. Total volume = ?		
29	Water tank $50 \times 30 \times 40$ cm, water depth 25 cm. Place object A (volume 9000 cm^3) first, then object B (volume 6000 cm^3). Final water depth = ? Did water spill out during the process? If so, how much overflow?		

30	The rectangular water tank is $30 \times 20 \times 15$ cm and the water depth is 10 cm. First remove 400 cm^3 of water and then add a stone (volume 700 cm^3). Final water depth = ?		
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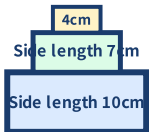
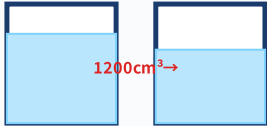
IV. application questions special topic (SSPA text question)

#	Question	Difficulty	Working Space (column → calculate → answer sentence)
31	An L-shaped planter consists of two parts: the bottom layer is 100 cm long, 40 cm wide and 30 cm high; the upper layer is 40 cm long, 40 cm wide and 50 cm high on the right side. What is the total volume of the planter in cm^3 ? How many m^3 is it?		
32	The internal length of a rectangular fish tank is 50 cm, the width is 30 cm, and the height is 40 cm. The original water depth is 25 cm. After placing a rockery (completely submerged), the water level rose to 32 cm. What is the volume of the rockery in cm^3 ?		
33	A rectangular water tank is $40 \times 25 \times 20$ cm with a water depth of 15 cm. A stone with a volume of 4000 cm^3 is placed. (a) Will the water spill? (b) If there is overflow, how many cm^3 is overflowed?		
34	The side length of the cube water tank is 20 cm, and the original water depth is 12 cm. Place a metal cube with a side length of 8 cm. (a) Volume of metal block = ? (b) How many cm did the water level rise? (c) Final water depth = ?		
35	The inside of a water tank is 60 cm long, 40 cm wide, and 30 cm high. There is water in the tank, the water depth is 20 cm. Put all 5 identical cube metal blocks (each side is 6 cm long) into the water. (a) Total volume of 5 metal blocks = ? (b) Will the water level overflow? (c) Final water depth = ?		

application question (continued)

#	Question	Difficulty	Working Space
36	A composite solid consists of a cuboid ($20 \times 10 \times 8$ cm) and a cube (side length 6 cm), with the cube attached to the center of the top of the cuboid. Total volume = ?		
37	Water tank $35 \times 25 \times 18$ cm, water depth 12 cm. After dropping an irregular stone block (completely submerged), the water depth becomes 16 cm. Volume of stone = ? If the stone is removed and an object with a volume of 2000 cm^3 is placed in it, will the water overflow?		

V. Ultimate challenge area

#	Question	Difficulty	Working Space
1	 Water tank $30 \times 20 \times 15$ cm. First inject some water to a depth of h cm. After placing a cube with a side length of 10 cm, the water level is just full (15 cm) without overflowing. Find the original water depth h.		
2	 A composite solid is formed by stacking three cubes (pyramid-shaped): the bottom cube has a side length of 10 cm, the middle cube has a side length of 7 cm, and the top cube has a side length of 4 cm, all stacked in the center. (a) Total volume = ? (b) If this composite solid is placed in a cubic tank filled with water (side length 15 cm, water depth 8 cm), will the water overflow? How much overflow? (assuming full submersion)		
3	 Tank A ($30 \times 20 \times 15$ cm, water depth 10 cm) and tank B ($25 \times 15 \times 12$ cm, water depth 7 cm). Take 1200 cm^3 water from water tank A and pour it into water tank B. (a) New water depth in tank A = ? (b) New water depth in tank B = ? (c) Is there any water overflowing from tank B?		

VI. Class after homework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
H1	L-shaped three-dimensional: bottom layer $10 \times 6 \times 4$ cm, upper right side $4 \times 6 \times 3$ cm. Total volume = ?		
H2	The water tank is $25 \times 12 \times 15$ cm, and the original water depth is 6 cm. The water level rose to 11 cm after placing the stones. Volume of stone = ?		
H3	Water tank $20 \times 15 \times 10$ cm, water depth 8 cm. Place an object with a volume of 500 cm^3 . Will the water overflow? If so, how much overflow?		
H4	Cuboid $18 \times 12 \times 10$ cm, with one corner cut off ($6 \times 4 \times 5$ cm). Remaining volume = ?		
H5	The side length of the cube water tank is 18 cm and the water depth is 12 cm. Place a cube of iron with a side length of 6 cm. Final water depth = ? (Accurate to 1 decimal		

place)

For advanced, choose doquestion (total 2 questions)

#	Question	Difficulty	Working Space
H6	Water tank $40 \times 20 \times 25$ cm, water depth 15 cm. First put in stone A (volume 3000 cm^3), then take out 1000 cm^3 of water, and finally put in stone B (volume 2000 cm^3). Final water depth = ?		
H7	An empty water tank $30 \times 20 \times 18$ cm. Pour 6 L of water ($1 \text{ L} = 1000 \text{ cm}^3$) and place a cube with a side of 8 cm. Final water depth = ? Will the water overflow?		

VII. The Lessoncorecommon errorssummary

Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	The composite solid is calculated as a complete cuboid SEP : The L shape is directly multiplied by the peripheral dimensions: L shape is directly multiplied by the peripheral dimensions	It must be divided into several simple solids, calculated separately and then summed.
2	Repeat calculation of overlapping parts after segmentation SEP : The overlapping area of two segmented blocks is calculated twice: The overlapping area of two divided blocks is counted twice.	The dividing line must be on the boundary to ensure that the blocks do not overlap each other
3	The wrong water level is used in the drainage method SEP : $V = \text{bottom area} \times \text{new water level (wrong!)}$: $V = \text{bottom area} \times \text{new water level (wrong!)}$	$V = \text{bottom area} \times (\text{new water level} - \text{original water level}) = \text{bottom area} \times \text{water level difference}$
4	Overflow issues are not checked SEP : Directly apply the drainage formula without checking whether the theoretical water level exceeds the height of the container: Apply the drainage formula directly without checking whether the theoretical water level exceeds the container height.	The first step is to calculate the available space; the second step is to determine whether there is overflow; the third step is to calculate the overflow amount.
5	Wrong base area SEP : The base area of the object is used instead of the base area of the container.: The bottom area of the object is used instead of the bottom area of the container.	The bottom area in the drainage method formula is The bottom area of the container SEP , not the object , not an object
6	Unit confusion: Forgot to convert between L (litres) and cm^3 ($1 \text{ L} = 1000 \text{ cm}^3$)	All volumes are first unified to cm^3 and then converted to the required units.

7	Multiple operations (adding water, taking water, placing objects) are in chaotic order	Calculate step by step in chronological order. Find the new water level at each step before proceeding to the next step.
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